

MULTIPLE LINEAR REGRESSION

Can age and experience explain income?

Use multiple linear regression to estimate income from two human variables.

DATA STORY / 01

Made with GAMMA

01 — INPUTS → PREDICTION

The project turns two inputs into one prediction



Age

Feature



Work experience

Feature



Annual or monthly income

Target

A small dataset makes every row matter

20

TOTAL ROWS

16

TRAINING ROWS

4

TESTING ROWS

A focused demonstration, not a universal income model.

The workflow moves from raw data to a fitted model



Load and inspect
the CSV



Separate
features from the target



Split, fit, predict, evaluate
from learned coefficients to evidence

Train on 80%, test on the unseen 20%

TRAINING SET · 80%

$(16, 2)$

Training shape: (16, 2)

The model learns from these rows.

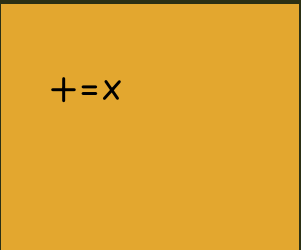
TESTING SET · 20%

$(4, 2)$

Testing shape: (4, 2)

The split checks whether the model generalizes beyond the rows it learned from.

The model's equation keeps the logic visible



$+=\times$

$$\text{Income} = 31465.06 - (101.06 \times \text{age}) + (2154.81 \times \text{experience})$$

INTERCEPT

31465.06

AGE COEFFICIENT

-101.06

EXPERIENCE COEFFICIENT

2154.81

06 — COEFFICIENTS

Experience is
the strongest
positive signal in
this fit

+\$2,154.81

PER ADDITIONAL YEAR OF EXPERIENCE

Holding age constant.

-101.06

AGE COEFFICIENT

The negative age coefficient reflects the specific pattern in this small dataset.

$R^2 = 0.94$: the model explains most observed variation

0.94

R-SQUARED

94% of variance explained

753,796.77

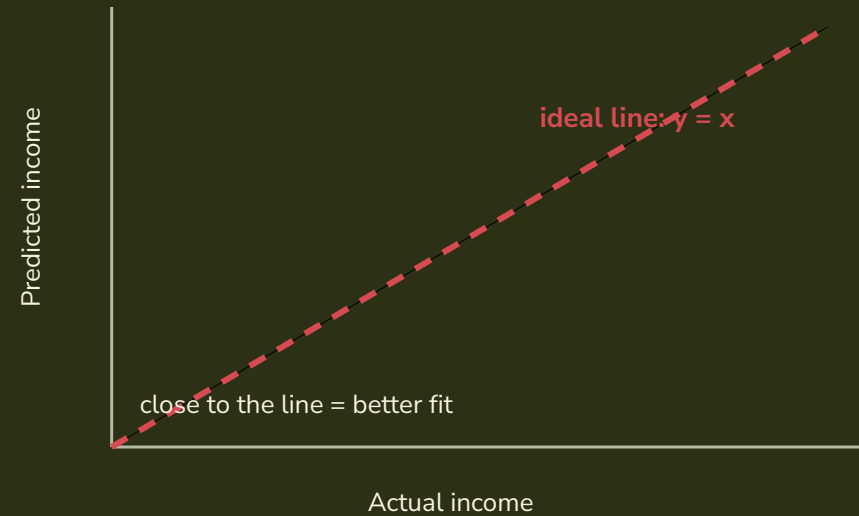
MEAN SQUARED ERROR

A second metric for reading the fit.

The headline metric pairs with a visual of actual versus predicted income.

Actual and predicted income sit close to the ideal line

Proximity to the line is the visual read on model fit.



Strong first result, clear next step

Within this project's small sample

The model performs well.

Build confidence

Improve confidence with a larger dataset.

Compare approaches

Random Forest or Gradient Boosting.